

Day-5- (02-01-2026) MongoDB

- MongoDB Shell Connected to Localhost

```
C:\Users\vrehe>mongosh
Current Mongosh Log ID: 69576161fd26fe9e511e2620
Connecting to:  mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.5.10
Using MongoDB:  8.2.3
Using Mongosh:  2.5.10

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

-----
The server generated these startup warnings when booting
2025-12-28T22:27:55.622+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----
```

- Existing Databases

```
test> show dbs
admin          40.00 KiB
backendDB      8.00 KiB
blogDB         8.00 KiB
config        36.00 KiB
local         72.00 KiB
```

CRUD Operations

- CREATE Operation

- Database and Collections creation

```
test> use InsuranceDB
switched to db InsuranceDB
InsuranceDB> db.createCollection("customers")
{ ok: 1 }
InsuranceDB> db.createCollection("policies")
{ ok: 1 }
InsuranceDB> db.createCollection("agents")
{ ok: 1 }
InsuranceDB> db.createCollection("policyAssignments")
{ ok: 1 }
InsuranceDB> db.createCollection("claims")
{ ok: 1 }
InsuranceDB> db.createCollection("payments")
{ ok: 1 }
```

- **Data Insertion into Collections**

Data Insertion into Customers Collection

```
/*Data Insertion*/
db.customers.insertMany([
{ CustomerID: 1, FirstName: "Rahul", LastName: "Sharma", DateOfBirth: "1990-05-12", Phone: "9876543210", Email: "rahul@gmail.com",
  City: "Mumbai" },
{ CustomerID: 2, FirstName: "Neha", LastName: "Verma", DateOfBirth: "1988-11-03", Phone: "9876501234", Email: "neha@gmail.com",
  City: "Delhi" },
{ CustomerID: 3, FirstName: "Amit", LastName: "Patel", DateOfBirth: "1995-02-21", Phone: "9988776655", Email: "amit@gmail.com",
  City: "Ahmedabad" },
{ CustomerID: 4, FirstName: "Priya", LastName: "Nair", DateOfBirth: "1992-07-09", Phone: "9776655443", Email: "priya@gmail.com",
  City: "Chennai" },
{ CustomerID: 5, FirstName: "Kiran", LastName: "Rao", DateOfBirth: "1985-04-15", Phone: "9123456789", Email: "kiran@gmail.com",
  City: "Hyderabad" },
{ CustomerID: 6, FirstName: "Vikas", LastName: "Singh", DateOfBirth: "1998-01-30", Phone: "9090909090", Email: "vikas@gmail.com",
  City: "Pune" },
{ CustomerID: 7, FirstName: "Anjali", LastName: "Kapoor", DateOfBirth: "1993-08-18", Phone: "9811122233", Email: "anjali@gmail.com",
  City: "Delhi" },
{ CustomerID: 8, FirstName: "Rohan", LastName: "Mehta", DateOfBirth: "1999-12-02", Phone: "9812345678", Email: "rohan@gmail.com",
  City: "Mumbai" },
{ CustomerID: 9, FirstName: "Sneha", LastName: "Pillai", DateOfBirth: "1991-03-13", Phone: "9000012345", Email: "sneha@gmail.com",
  City: "Chennai" },
{ CustomerID: 10, FirstName: "Arjun", LastName: "Desai", DateOfBirth: "1987-06-25", Phone: "9555577777", Email: "arjun@gmail.com",
  City: "Bangalore" }
])
```

Data Insertion into Policies Collection

```
db.policies.insertMany([
{ PolicyID: 101, PolicyName: "Health Secure", PolicyType: "Health", PremiumAmount: 30000, DurationYears: 5 },
{ PolicyID: 102, PolicyName: "Life Shield", PolicyType: "Life", PremiumAmount: 45000, DurationYears: 10 },
{ PolicyID: 103, PolicyName: "Car Protect", PolicyType: "Vehicle", PremiumAmount: 25000, DurationYears: 3 },
{ PolicyID: 104, PolicyName: "Family Health", PolicyType: "Health", PremiumAmount: 40000, DurationYears: 7 },
{ PolicyID: 105, PolicyName: "Senior Life", PolicyType: "Life", PremiumAmount: 55000, DurationYears: 12 },
{ PolicyID: 106, PolicyName: "Bike Secure", PolicyType: "Vehicle", PremiumAmount: 18000, DurationYears: 2 },
{ PolicyID: 107, PolicyName: "Term Plus", PolicyType: "Life", PremiumAmount: 50000, DurationYears: 15 },
{ PolicyID: 108, PolicyName: "MediCare", PolicyType: "Health", PremiumAmount: 32000, DurationYears: 6 },
{ PolicyID: 109, PolicyName: "Travel Guard", PolicyType: "Travel", PremiumAmount: 15000, DurationYears: 1 },
{ PolicyID: 110, PolicyName: "Child Future", PolicyType: "Life", PremiumAmount: 48000, DurationYears: 12 }
])
```

Data Insertion into Agents Collection

```
db.agents.insertMany([
{ AgentID: 201, AgentName: "Suresh Kumar", Phone: "8888881111", City: "Mumbai" },
{ AgentID: 202, AgentName: "Rita Mehra", Phone: "8888882222", City: "Delhi" },
{ AgentID: 203, AgentName: "Mahesh Iyer", Phone: "8888883333", City: "Chennai" },
{ AgentID: 204, AgentName: "Anita Shah", Phone: "8888884444", City: "Ahmedabad" },
{ AgentID: 205, AgentName: "Deepak Singh", Phone: "8888885555", City: "Pune" },
{ AgentID: 206, AgentName: "Kavita Rao", Phone: "8888886666", City: "Hyderabad" },
{ AgentID: 207, AgentName: "Vivek Patel", Phone: "8888887777", City: "Mumbai" },
{ AgentID: 208, AgentName: "Sunita Nair", Phone: "8888889999", City: "Chennai" },
{ AgentID: 209, AgentName: "Aman Gupta", Phone: "7777771111", City: "Delhi" },
{ AgentID: 210, AgentName: "Pooja Deshmukh", Phone: "7777772222", City: "Pune" }
])
```

Data Insertion into PolicyAssignments Collection

```
db.policyAssignments.insertMany([
  { AssignmentID: 301, CustomerID: 1, PolicyID: 101, AgentID: 201, StartDate: "2024-01-01", EndDate: "2029-01-01" },
  { AssignmentID: 302, CustomerID: 2, PolicyID: 102, AgentID: 202, StartDate: "2024-03-01", EndDate: "2034-03-01" },
  { AssignmentID: 303, CustomerID: 3, PolicyID: 103, AgentID: 203, StartDate: "2023-05-10", EndDate: "2026-05-10" },
  { AssignmentID: 304, CustomerID: 4, PolicyID: 104, AgentID: 204, StartDate: "2024-02-12", EndDate: "2031-02-12" },
  { AssignmentID: 305, CustomerID: 5, PolicyID: 105, AgentID: 205, StartDate: "2023-11-01", EndDate: "2035-11-01" },
  { AssignmentID: 306, CustomerID: 6, PolicyID: 106, AgentID: 206, StartDate: "2024-06-15", EndDate: "2026-06-15" },
  { AssignmentID: 307, CustomerID: 7, PolicyID: 107, AgentID: 207, StartDate: "2024-07-07", EndDate: "2039-07-07" },
  { AssignmentID: 308, CustomerID: 8, PolicyID: 108, AgentID: 208, StartDate: "2024-08-20", EndDate: "2030-08-20" },
  { AssignmentID: 309, CustomerID: 9, PolicyID: 109, AgentID: 209, StartDate: "2024-09-10", EndDate: "2025-09-10" },
  { AssignmentID: 310, CustomerID: 10, PolicyID: 110, AgentID: 210, StartDate: "2024-10-01", EndDate: "2036-10-01" }
])
```

Data Insertion into Claims Collection

```
db.claims.insertMany([
  { ClaimID: 401, AssignmentID: 301, ClaimDate: "2024-04-15", ClaimAmount: 20000, ClaimStatus: "Approved" },
  { ClaimID: 402, AssignmentID: 302, ClaimDate: "2024-06-01", ClaimAmount: 50000, ClaimStatus: "Pending" },
  { ClaimID: 403, AssignmentID: 303, ClaimDate: "2023-12-20", ClaimAmount: 15000, ClaimStatus: "Rejected" },
  { ClaimID: 404, AssignmentID: 304, ClaimDate: "2024-03-05", ClaimAmount: 30000, ClaimStatus: "Approved" },
  { ClaimID: 405, AssignmentID: 305, ClaimDate: "2024-01-25", ClaimAmount: 60000, ClaimStatus: "Approved" },
  { ClaimID: 406, AssignmentID: 306, ClaimDate: "2024-05-19", ClaimAmount: 12000, ClaimStatus: "Pending" },
  { ClaimID: 407, AssignmentID: 307, ClaimDate: "2024-08-10", ClaimAmount: 75000, ClaimStatus: "Approved" },
  { ClaimID: 408, AssignmentID: 308, ClaimDate: "2024-09-11", ClaimAmount: 18000, ClaimStatus: "Rejected" },
  { ClaimID: 409, AssignmentID: 309, ClaimDate: "2024-10-02", ClaimAmount: 9000, ClaimStatus: "Pending" },
  { ClaimID: 410, AssignmentID: 310, ClaimDate: "2024-11-15", ClaimAmount: 65000, ClaimStatus: "Approved" }
])
```

Data Insertion into Payments Collection

```
db.payments.insertMany([
  { PaymentID: 501, PolicyID: 101, PaymentDate: "2024-01-10", Amount: 30000 },
  { PaymentID: 502, PolicyID: 102, PaymentDate: "2024-03-10", Amount: 45000 },
  { PaymentID: 503, PolicyID: 103, PaymentDate: "2023-05-15", Amount: 25000 },
  { PaymentID: 504, PolicyID: 104, PaymentDate: "2024-02-20", Amount: 40000 },
  { PaymentID: 505, PolicyID: 105, PaymentDate: "2023-11-10", Amount: 55000 },
  { PaymentID: 506, PolicyID: 106, PaymentDate: "2024-06-20", Amount: 18000 },
  { PaymentID: 507, PolicyID: 107, PaymentDate: "2024-07-15", Amount: 50000 },
  { PaymentID: 508, PolicyID: 108, PaymentDate: "2024-08-25", Amount: 32000 },
  { PaymentID: 509, PolicyID: 109, PaymentDate: "2024-09-15", Amount: 15000 },
  { PaymentID: 510, PolicyID: 110, PaymentDate: "2024-10-20", Amount: 48000 }
])
```

READ Operation

Reading all the Data in each collection

```
db.customers.find()
db.policies.find()
db.agents.find()
db.policyAssignments.find()
db.claims.find()
db.payments.find()
```

MongoDB search & analytical queries

1. Find all policies of type “Health”

```
db.policies.find({ PolicyType: "Health" })
```

Output:

```
[
  {
    "_id": {
      "$oid": "695768e4050b01ef46ce662f"
    },
    "PolicyID": 101,
    "PolicyName": "Health Secure",
    "PolicyType": "Health",
    "PremiumAmount": 30000,
    "DurationYears": 5
  },
  {
    "_id": {
      "$oid": "695768e4050b01ef46ce6632"
    },
    "PolicyID": 104,
    "PolicyName": "Family Health",
    "PolicyType": "Health",
    "PremiumAmount": 40000,
    "DurationYears": 7
  },
  {
    "_id": {
      "$oid": "695768e4050b01ef46ce6636"
    },
    "PolicyID": 108,
    "PolicyName": "MediCare",
    "PolicyType": "Health",
    "PremiumAmount": 32000,
    "DurationYears": 6
  }
]
```

2. Find customers living in Delhi

```
db.customers.find({ City: "Delhi" })
```

Output:

```
[
  {
    "_id": {
      "$oid": "695768bd51ee5307cb7d8619"
    },
    "CustomerID": 2,
    "FirstName": "Neha",
    "LastName": "Verma",
    "DateOfBirth": "1988-11-03",
    "Phone": "9876501234",
    "Email": "neha@gmail.com",
    "City": "Delhi"
  },
  {
    "_id": {
      "$oid": "695768bd51ee5307cb7d861e"
    },
    "CustomerID": 7,
    "FirstName": "Anjali",
    "LastName": "Kapoor",
    "DateOfBirth": "1993-08-18",
    "Phone": "9811122233",
    "Email": "anjali@gmail.com",
    "City": "Delhi"
  }
]
```

3. Search customers whose first name starts with 'A'

```
db.customers.find({ FirstName: { $regex: '^A', $options: 'i' } })
```

Output:

```
[
  {
    "_id": {
      "$oid": "695768bd51ee5307cb7d861a"
    },
    "CustomerID": 3,
    "FirstName": "Amit",
    "LastName": "Patel",
    "DateOfBirth": "1995-02-21",
    "Phone": "9988776655",
    "Email": "amit@gmail.com",
    "City": "Ahmedabad"
  },
  {
    "_id": {
      "$oid": "695768bd51ee5307cb7d861e"
    },
    "CustomerID": 7,
    "FirstName": "Anjali",
    "LastName": "Kapoor",
    "DateOfBirth": "1993-08-18",
    "Phone": "9811122233",
    "Email": "anjali@gmail.com",
    "City": "Delhi"
  },
  {
    "_id": {
      "$oid": "695768bd51ee5307cb7d8621"
    },
    "CustomerID": 10,
    "FirstName": "Arjun",
    "LastName": "Desai",
    "DateOfBirth": "1987-06-25",
    "Phone": "9555577777",
    "Email": "arjun@gmail.com",
    "City": "Bangalore"
  }
]
```

4. Get claims which are Rejected

```
db.claims.find({ ClaimStatus: "Rejected" })
```

Output:

```
[
  {
    "_id": {
      "$oid": "69576911882cbffdb6329cc8"
    },
    "ClaimID": 403,
    "AssignmentID": 303,
    "ClaimDate": "2023-12-20",
    "ClaimAmount": 15000,
    "ClaimStatus": "Rejected"
  },
  {
    "_id": {
      "$oid": "69576911882cbffdb6329ccd"
    },
    "ClaimID": 408,
    "AssignmentID": 308,
    "ClaimDate": "2024-09-11",
    "ClaimAmount": 18000,
    "ClaimStatus": "Rejected"
  }
]
```

5. Policies with Premium more than 49,999

```
db.policies.find({ PremiumAmount: { $gt: 49999 } })
```

Output:

```
[
  {
    Edit Document
    "_id": {
      "$oid": "695768e4050b01ef46ce6633"
    },
    "PolicyID": 105,
    "PolicyName": "Senior Life",
    "PolicyType": "Life",
    "PremiumAmount": 55000,
    "DurationYears": 12
  },
  {
    Edit Document
    "_id": {
      "$oid": "695768e4050b01ef46ce6635"
    },
    "PolicyID": 107,
    "PolicyName": "Term Plus",
    "PolicyType": "Life",
    "PremiumAmount": 50000,
    "DurationYears": 15
  }
]
```

- UPDATE Operation
- Updating Data in collections

1. Update the phone number of the customer with id 1.

Updation

```
db.customers.updateOne(  
  { CustomerID: 1 },  
  { $set: { Phone: "9999999999" } }  
)
```

After Updation

```
[  
  {  
    "Phone": "9876999990"  
  }  
]
```

2. Change a customer city

Updation

```
db.customers.updateOne(  
  { CustomerID: 5 },  
  { $set: { City: "Kolkata" } }  
)
```

After Updation

```
[  
  {  
    "city": "Kolkata"  
  }  
]
```

3. Increase premium of a specific policy

Updation

```
db.policies.updateOne(  
  { PolicyID: 102 },  
  { $inc: { PremiumAmount: 50000 } }  
)
```

After Updation

```
[  
  {  
    "PremiumAmount": 95000  
  }  
)
```

4. Update claim status to Approved

Updation

```
db.claims.updateOne(  
  { ClaimID: 402 },  
  { $set: { ClaimStatus: "Approved" } }  
)
```

After Updation

```
[  
  {  
    "ClaimStatus": "Approved"  
  }  
)
```

- DELETE Operation

- Deleting One Data in collections

1. Delete one customer with id 5

```
db.customers.deleteOne({ CustomerID: 5 })
```

2. Delete one policy assignment with id 309

```
db.policyAssignments.deleteOne({ AssignmentID: 309 })
```

3. Delete one payment with id 509

```
db.payments.deleteOne({ PaymentID: 509 })
```

- Deleting Many Data in collections

4. Delete all customers from Delhi

```
db.customers.deleteMany({ City: "Delhi" })
```

5. Delete all rejected claims

```
db.claims.deleteMany({ ClaimStatus: "Rejected" })
```

6. Delete payments older than a date

```
db.payments.deleteMany({  
  PaymentDate: { $lt: "2024-01-01" }  
})
```

7. Delete all expired policy assignments

```
db.policyAssignments.deleteMany({  
  EndDate: { $lt: "2025-01-01" }  
})
```

8. Delete Pending Claims older than 1 year

```
db.claims.deleteMany({  
  ClaimStatus: "Pending",  
  ClaimDate: { $lt: "2024-01-01" }  
})
```

Some Queries

1. Show only policyName and policyType

```
db.policies.find(  
  {},{_id:0,PolicyName:1,PolicyType:1}  
)
```

Output:

```
[  
  {  
    "PolicyName": "Health Secure",  
    "PolicyType": "Health"  
  },  
  {  
    "PolicyName": "Life Shield",  
    "PolicyType": "Life"  
  },  
  {  
    "PolicyName": "Car Protect",  
    "PolicyType": "Vehicle"  
  },  
  {  
    "PolicyName": "Family Health",  
    "PolicyType": "Health"  
  },  
  {  
    "PolicyName": "Senior Life",  
    "PolicyType": "Life"  
  },  
  {  
    "PolicyName": "Bike Secure",  
    "PolicyType": "Vehicle"  
  },  
  {  
    "PolicyName": "Term Plus",  
    "PolicyType": "Life"  
  },  
  {  
    "PolicyName": "MediCare",  
    "PolicyType": "Health"  
  },  
  {  
    "PolicyName": "Travel Guard",  
    "PolicyType": "Travel"  
  },  
]
```

2. Exclude `_id` in Customers Collection

```
db.customers.find(  
  {},{_id:0}  
)
```

Output:

```
[  
  {  
    "PolicyID": 101,  
    "PolicyName": "Health Secure",  
    "PolicyType": "Health",  
    "PremiumAmount": 30000,  
    "DurationYears": 5  
  },  
  {  
    "PolicyID": 102,  
    "PolicyName": "Life Shield",  
    "PolicyType": "Life",  
    "PremiumAmount": 95000,  
    "DurationYears": 10  
  },  
  {  
    "PolicyID": 103,  
    "PolicyName": "Car Protect",  
    "PolicyType": "Vehicle",  
    "PremiumAmount": 25000,  
    "DurationYears": 3  
  },  
  {  
    "PolicyID": 104,  
    "PolicyName": "Family Health",  
    "PolicyType": "Health",  
    "PremiumAmount": 40000,  
    "DurationYears": 7  
  },  
  {  
    "PolicyID": 105,  
    "PolicyName": "Senior Life",  
    "PolicyType": "Life",  
    "PremiumAmount": 55000,  
    "DurationYears": 12  
  },  
]
```

3. Show all fields except premium

```
db.policies.find(
  {}, {PremiumAmount:0}
)
```

Output:

```
[
  {
    "_id": {
      "$oid": "695768e4050b01ef46ce662f"
    },
    "PolicyID": 101,
    "PolicyName": "Health Secure",
    "PolicyType": "Health",
    "DurationYears": 5
  },
  {
    "_id": {
      "$oid": "695768e4050b01ef46ce6630"
    },
    "PolicyID": 102,
    "PolicyName": "Life Shield",
    "PolicyType": "Life",
    "DurationYears": 10
  },
  {
    "_id": {
      "$oid": "695768e4050b01ef46ce6631"
    },
    "PolicyID": 103,
    "PolicyName": "Car Protect",
    "PolicyType": "Vehicle",
    "DurationYears": 3
  },
  {
    "_id": {
      "$oid": "695768e4050b01ef46ce6632"
    },
    "PolicyID": 104,
    "PolicyName": "Family Health",
    "PolicyType": "Health",
    "DurationYears": 7
  },
]
```

4. Mark all claims after June,2025 as pending

Updation:

```
db.claims.updateMany(  
  { ClaimDate: { $gt: "2024-06-30" } },  
  { $set: { ClaimStatus: "Pending" } }  
)
```

After Updation:

```
db.claims.updateMany(  
  { ClaimDate: { $gt: "2024-06-30" } },  
  { $set: { ClaimStatus: "Pending" } }  
)
```

5. Customers NOT from Delhi

```
db.customers.find({ city: { $ne: "Delhi" } })
```

Output:

```
[  
  {  
    "_id": {  
      "$oid": "695768bd51ee5307cb7d8618"  
    },  
    "CustomerID": 1,  
    "FirstName": "Rahul",  
    "LastName": "Sharma",  
    "DateOfBirth": "1990-05-12",  
    "Phone": "9876999990",  
    "Email": "rahul@gmail.com",  
    "City": "Mumbai"  
  },  
  {  
    "_id": {  
      "$oid": "695768bd51ee5307cb7d8619"  
    },  
    "CustomerID": 2,  
    "FirstName": "Neha",  
    "LastName": "Verma",  
    "DateOfBirth": "1988-11-03",  
    "Phone": "9876501234",  
    "Email": "neha@gmail.com",  
    "City": "Delhi"  
  },  
  {  
    "_id": {  
      "$oid": "695768bd51ee5307cb7d861a"  
    },  
    "CustomerID": 3,  
    "FirstName": "Amit",  
    "LastName": "Patel",  
    "DateOfBirth": "1995-02-21",  
    "Phone": "9988776655",  
    "Email": "amit@gmail.com",  
    "City": "Ahmedabad"  
  },  
]
```

6. Policies where premium is NOT between 10k–20k

```
db.policies.find({
  PremiumAmount: { $not: { $gte: 10000, $lte: 20000 } }
})
```

Output:

```
[
  {
    "_id": {
      "$oid": "695768e4050b01ef46ce662f"
    },
    "PolicyID": 101,
    "PolicyName": "Health Secure",
    "PolicyType": "Health",
    "PremiumAmount": 30000,
    "DurationYears": 5
  },
  {
    "_id": {
      "$oid": "695768e4050b01ef46ce6630"
    },
    "PolicyID": 102,
    "PolicyName": "Life Shield",
    "PolicyType": "Life",
    "PremiumAmount": 95000,
    "DurationYears": 10
  },
  {
    "_id": {
      "$oid": "695768e4050b01ef46ce6631"
    },
    "PolicyID": 103,
    "PolicyName": "Car Protect",
    "PolicyType": "Vehicle",
    "PremiumAmount": 25000,
    "DurationYears": 3
  },
]
```

7. Find top 2 Premium Policies

```
db.policies.find(  
  {},  
  { _id: 0, PolicyName: 1, PremiumAmount: 1 }  
)  
.sort({ PremiumAmount: -1 })  
.limit(2)
```

Output:

```
[  
  {  
    "PolicyName": "Life Shield",  
    "PremiumAmount": 95000  
  },  
  {  
    "PolicyName": "Senior Life",  
    "PremiumAmount": 55000  
  }  
]
```

8. Find the policy with the lowest premium amount

```
db.policies.find(  
  {},  
  { _id: 0, PolicyName: 1, PremiumAmount: 1 }  
)  
.sort({ PremiumAmount: 1 })  
.limit(1)
```

Output:

```
[  
  {  
    "PolicyName": "Travel Guard",  
    "PremiumAmount": 15000  
  }  
]
```

Aggregation Queries

1. Count number of policies by Policy Type

```
db.policies.aggregate([
  {
    $group: {
      _id: "$PolicyType",
      TotalPolicies: { $sum: 1 }
    }
  }
])
```

Output:

```
[
  {
    "_id": "Life",
    "TotalPolicies": 4
  },
  {
    "_id": "Health",
    "TotalPolicies": 3
  },
  {
    "_id": "Vehicle",
    "TotalPolicies": 2
  },
  {
    "_id": "Travel",
    "TotalPolicies": 1
  }
]
```

2. Total Claim Amount by Claim Status

```
db.claims.aggregate([
  {
    $group: {
      _id: "$ClaimStatus",
      TotalClaimAmount: { $sum: "$ClaimAmount" }
    }
  }
])
```

Output:

```
[
  {
    "_id": "Pending",
    "TotalClaimAmount": 179000
  },
  {
    "_id": "Approved",
    "TotalClaimAmount": 160000
  },
  {
    "_id": "Rejected",
    "TotalClaimAmount": 15000
  }
]
```

3. Total payments received per Policy

```
db.payments.aggregate([
  {
    $group: {
      _id: "$PolicyID",
      TotalPaid: { $sum: "$Amount" },
      PaymentsCount: { $sum: 1 }
    }
  }
])
```

Output:

```
[
  {
    "_id": 104,
    "TotalPaid": 40000,
    "PaymentsCount": 1
  },
  {
    "_id": 103,
    "TotalPaid": 25000,
    "PaymentsCount": 1
  },
  {
    "_id": 101,
    "TotalPaid": 30000,
    "PaymentsCount": 1
  },
  {
    "_id": 102,
    "TotalPaid": 45000,
    "PaymentsCount": 1
  },
  {
    "_id": 106,
    "TotalPaid": 18000,
    "PaymentsCount": 1
  },
  {
    "_id": 105,
    "TotalPaid": 55000,
    "PaymentsCount": 1
  },
  {
    "_id": 107,
    "TotalPaid": 50000,
    "PaymentsCount": 1
  },
]
```

4. Count claims by status

```
db.claims.aggregate([
  {
    $group: {
      _id: "$ClaimStatus",
      TotalClaims: { $sum: 1 }
    }
  }
])
```

Output:

```
[
  {
    "_id": "Pending",
    "TotalClaims": 5
  },
  {
    "_id": "Approved",
    "TotalClaims": 4
  },
  {
    "_id": "Rejected",
    "TotalClaims": 1
  }
]
```

5. Number of Customers by City

```
db.customers.aggregate([
  {
    $group: {
      _id: "$City",
      TotalCustomers: { $sum: 1 }
    }
  }
])
```

Output:

```
[
  {
    "_id": "Chennai",
    "TotalCustomers": 2
  },
  {
    "_id": "Ahmedabad",
    "TotalCustomers": 1
  },
  {
    "_id": "Mumbai",
    "TotalCustomers": 2
  },
  {
    "_id": "Delhi",
    "TotalCustomers": 2
  },
  {
    "_id": "Pune",
    "TotalCustomers": 1
  },
  {
    "_id": "Bangalore",
    "TotalCustomers": 1
  }
]
```

Joins Like Features in MongoDB

1. Customers who have policies

```
db.policyAssignments.aggregate([
  {
    $lookup: {
      from: "customers",
      localField: "CustomerID",
      foreignField: "CustomerID",
      as: "Customer"
    }
  },
  { $unwind: "$Customer" },
  {
    $lookup: {
      from: "policies",
      localField: "PolicyID",
      foreignField: "PolicyID",
      as: "Policy"
    }
  },
  { $unwind: "$Policy" },
  {
    $project: {
      _id: 0,
      CustomerName: "$Customer.FirstName",
      PolicyName: "$Policy.PolicyName",
      PolicyType: "$Policy.PolicyType"
    }
  }
])
```

Output:

```
[
  {
    "CustomerName": "Rahul",
    "PolicyName": "Health Secure",
    "PolicyType": "Health"
  },
  {
    "CustomerName": "Neha",
    "PolicyName": "Life Shield",
    "PolicyType": "Life"
  },
  {
    "CustomerName": "Amit",
    "PolicyName": "Car Protect",
    "PolicyType": "Vehicle"
  },
  {
    "CustomerName": "Priya",
    "PolicyName": "Family Health",
    "PolicyType": "Health"
  },
  {
    "CustomerName": "Vikas",
    "PolicyName": "Bike Secure",
    "PolicyType": "Vehicle"
  }
]
```

2. Agents and Customers They Handle.

```
db.policyAssignments.aggregate([
  {
    $lookup: {
      from: "agents",
      localField: "AgentID",
      foreignField: "AgentID",
      as: "Agent"
    }
  },
  { $unwind: "$Agent" },
  {
    $lookup: {
      from: "customers",
      localField: "CustomerID",
      foreignField: "CustomerID",
      as: "Customer"
    }
  },
  { $unwind: "$Customer" },
  {
    $project: {
      _id: 0,
      AgentName: "$Agent.AgentName",
      CustomerName: "$Customer.FirstName"
    }
  }
])
```

Output:

```
[
  {
    "AgentName": "Suresh Kumar",
    "CustomerName": "Rahul"
  },
  {
    "AgentName": "Rita Mehra",
    "CustomerName": "Neha"
  },
  {
    "AgentName": "Mahesh Iyer",
    "CustomerName": "Amit"
  },
  {
    "AgentName": "Anita Shah",
    "CustomerName": "Priya"
  },
  {
    "AgentName": "Kavita Rao",
    "CustomerName": "Vikas"
  },
]
```

3. Claims with Customer Names

```
db.claims.aggregate([
  {
    $lookup: {
      from: "policyAssignments",
      localField: "AssignmentID",
      foreignField: "AssignmentID",
      as: "Assign"
    }
  },
  { $unwind: "$Assign" },
  {
    $lookup: {
      from: "customers",
      localField: "Assign.CustomerID",
      foreignField: "CustomerID",
      as: "Customer"
    }
  },
  { $unwind: "$Customer" },
  {
    $project: {
      _id: 0,
      CustomerName: "$Customer.FirstName",
      ClaimAmount: 1,
      ClaimStatus: 1
    }
  }
])
```

Output:

```
[
  {
    "ClaimAmount": 20000,
    "ClaimStatus": "Approved",
    "CustomerName": "Rahul"
  },
  {
    "ClaimAmount": 50000,
    "ClaimStatus": "Approved",
    "CustomerName": "Neha"
  },
  {
    "ClaimAmount": 15000,
    "ClaimStatus": "Rejected",
    "CustomerName": "Amit"
  },
  {
    "ClaimAmount": 30000,
    "ClaimStatus": "Approved",
    "CustomerName": "Priya"
  },
]
```

4. Policies with Assigned Agent

```
db.policyAssignments.aggregate([
  {
    $lookup: {
      from: "policies",
      localField: "PolicyID",
      foreignField: "PolicyID",
      as: "Policy"
    }
  },
  { $unwind: "$Policy" },
  {
    $lookup: {
      from: "agents",
      localField: "AgentID",
      foreignField: "AgentID",
      as: "Agent"
    }
  },
  { $unwind: "$Agent" },
  {
    $project: {
      _id: 0,
      PolicyName: "$Policy.PolicyName",
      AgentName: "$Agent.AgentName"
    }
  }
])
```

Output:

```
[
  {
    "PolicyName": "Health Secure",
    "AgentName": "Suresh Kumar"
  },
  {
    "PolicyName": "Life Shield",
    "AgentName": "Rita Mehra"
  },
  {
    "PolicyName": "Car Protect",
    "AgentName": "Mahesh Iyer"
  },
  {
    "PolicyName": "Family Health",
    "AgentName": "Anita Shah"
  },
]
```

5. Claims with Their Policy Name

```
db.claims.aggregate([
  {
    $lookup: {
      from: "policyAssignments",
      localField: "AssignmentID",
      foreignField: "AssignmentID",
      as: "Assign"
    }
  },
  { $unwind: "$Assign" },
  {
    $lookup: {
      from: "policies",
      localField: "Assign.PolicyID",
      foreignField: "PolicyID",
      as: "Policy"
    }
  },
  { $unwind: "$Policy" },
  {
    $project: {
      _id: 0,
      ClaimID: 1,
      ClaimAmount: 1,
      ClaimStatus: 1,
      PolicyName: "$Policy.PolicyName"
    }
  }
])
```

Output:

```
[
  {
    "ClaimID": 401,
    "ClaimAmount": 20000,
    "ClaimStatus": "Approved",
    "PolicyName": "Health Secure"
  },
  {
    "ClaimID": 402,
    "ClaimAmount": 50000,
    "ClaimStatus": "Approved",
    "PolicyName": "Life Shield"
  },
  {
    "ClaimID": 403,
    "ClaimAmount": 15000,
    "ClaimStatus": "Rejected",
    "PolicyName": "Car Protect"
  },
]
```